AUSTIN TONSAY

U.S. Citizen | +1 (562) 213-6508 | austintonsay@gmail.com | linkedin.com/in/austintonsay | austintonsay.github.io

EDUCATION:

California State Polytechnic University, Pomona *Expected Graduation: May 2029*

Bachelor of Science, Aerospace Engineering (Emphasis: Astronautics) | GPA: 3.83

PERSONAL PROJECTS:

Baseline vs. Altitude-Optimized High-Power Rocketry Experiment: *May 2026 – September 2026*

- Designed, built, and launched baseline and altitude-optimized rockets in SolidWorks/OpenRocket to analyze effect of geometry changes on apogee while maintaining constant body diameter and motor; 3,136 ft and 4,131 ft apogees, respectively (31.7% altitude difference)
- Owned slicing and material optimization of 3D-printed PETG-CF and ABS parts to increase fin structural rigidity against flutter and prevent thermal fusion between motor and retention
- Developed 6-in interchangeable avionics bay to reduce electronics cost by $65
- Iterated 5 motor retention concepts to develop bayonet-style retainer and proved functionality in flight, earning L1 certification

EXPERIENCE:

AIAA | Design, Build, Fly (DBF) Competition: *August 2025 – August 2026*

Structures/Manufacturing Engineer

- Designed tail assembly in SolidWorks intended for 3D printing to reduce part count, enhance manufacturability, and improve aerodynamic performance from previous year.
- Completed 3 material trade studies comparing 4–6 candidates across bending/torsional strength, mass, cost, and manufacturability to guide wing/fuselage structural material selection.
- Owned wing/fuselage assembly, foam core construction with carbon-fiber layups, and fuselage surface finishing with Monokote heat-shrink covering to reduce drag.

Bronco S.T.A.R. | BLADE Odyssey: *August 2025 – May 2026*

Communications and Thermals Engineer

- Developed a heater trade study comparing 3 candidate heating solutions against a ~9.25 W / 37.2 Wh CubeSat power constraint, balancing thermal capability, power draw, size, and integration risk.
- Sized two 60 × 40 mm heating pads at 4.2 V, 3 W each (~0.71 A) within SWaP (size, weight, and power) constraints.
- Created thermostat-control pseudocode to cycle battery heating between 0–10°C with a 30%-of-flight runtime failsafe for ~4-hour CubeSat mission to maintain safe operating temperature for internal components.

Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA): *September 2025 – November 2025*

Small Rocket Competition, Team Lead

- Optimized model rocket fin and body geometries in OpenRocket, SolidWorks, and Bambu Studio to improve projected apogee from 2,100 ft to 2,900 ft (~38% increase).
- Led 5-person design/build team through aerodynamic design, manufacturing, and avionics integration while maintaining project schedule within 2 days of planned milestones.
- Programmed Arduino IDE for avionics systems to read rocket altitude, surrounding pressure, and temperature; post-flight failure analysis identified insufficient epoxy cure time as the cause of internal structural separation.

ACADEMIC PROJECTS:

ARO1021L - Analyzed OpenRocket, static-fire, inclinometer, and onboard altimeter apogee estimates across 3 model rockets; quantified method discrepancies and evaluated drag, manufacturing, and environmental sources of error.

EGR1000L - Drafted and programmed an Arduino-controlled animatronic system integrating 4 servos and an ~85% 3D-printed structure; iterated actuator selection and mechanical fit to achieve 3 coordinated motions within a 12-in design envelope.

SKILLS:

SolidWorks, Ansys Fluent, OpenRocket, Arduino IDE, PrusaSlicer, Computer-Aided Design (CAD), Computational Fluid Dynamics (CFD), FDM Additive Manufacturing, Carbon Fiber Layups, Wood/Foam Airframe Fabrication, Excel VBA